

PRAKTIKUM SISTEM OPERASI

MODUL 9
SYSCALL XINU



Oleh:

Dimastian Aji Wibowo

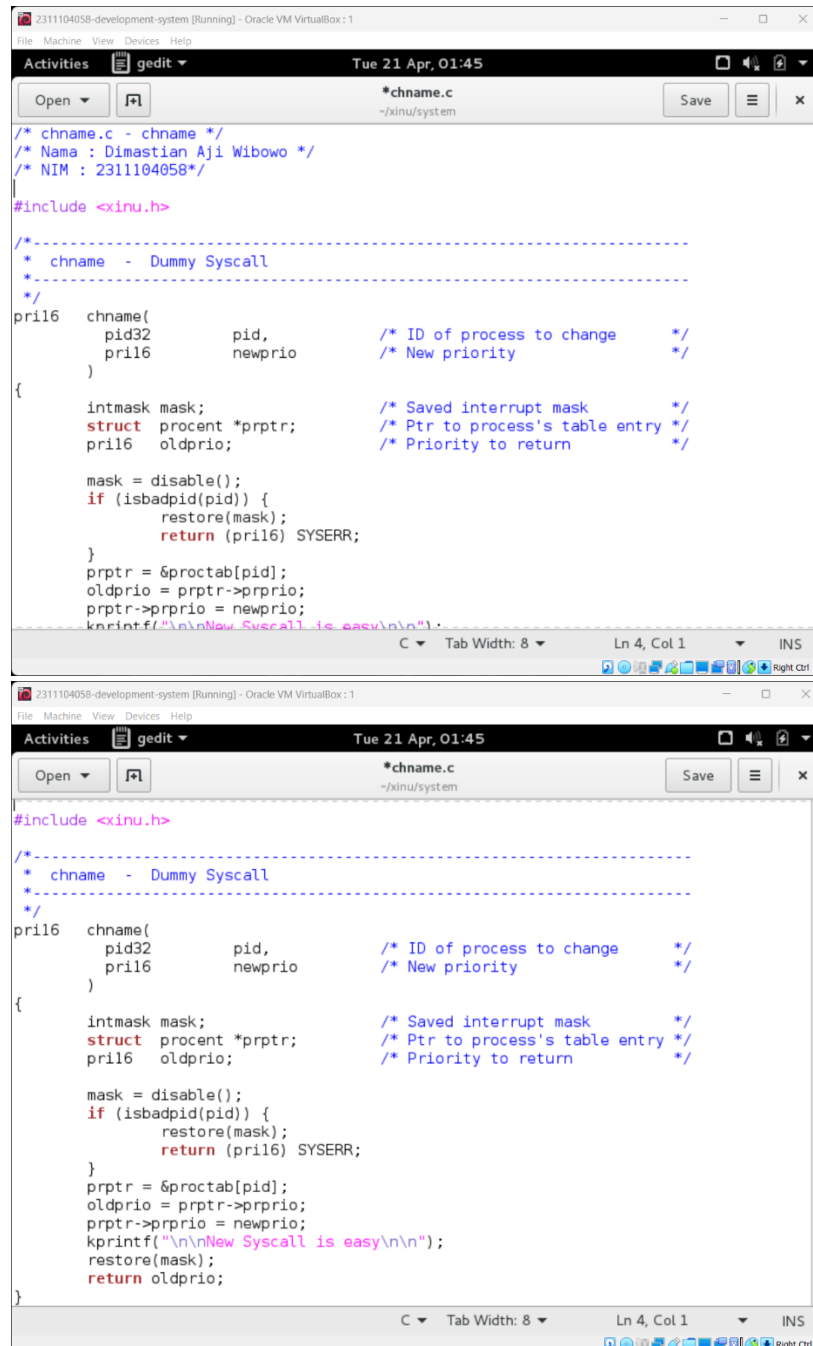
2311104058

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
DIREKTORAT KAMPUS PURWOKERTO
UNIVERSITAS TELKOM
2026

JURNAL

1. Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5!

File chname.c



```
/* chname.c - chname */
/* Nama : Dimastian Aji Wibowo */
/* NIM : 2311104058 */

#include <xinu.h>

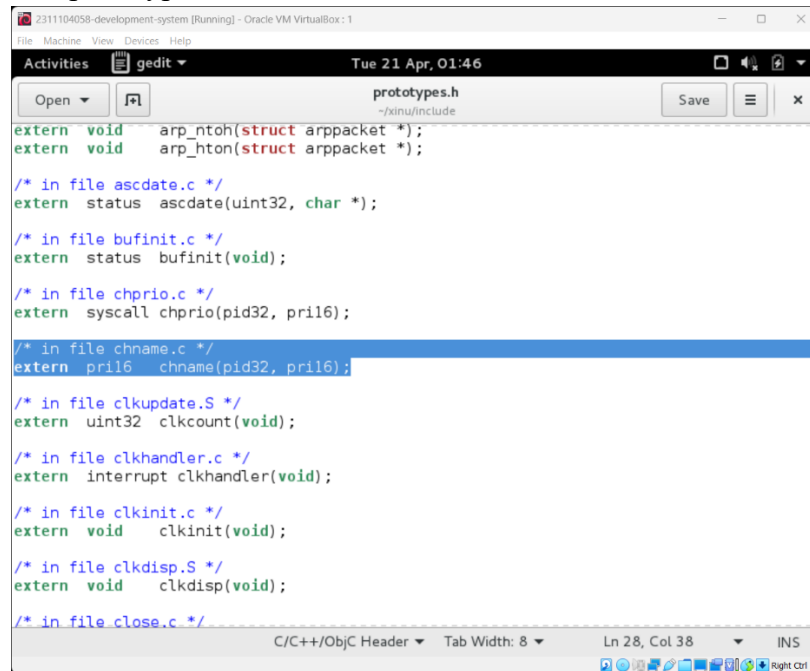
/*-----
 * chname - Dummy Syscall
 *-----
 */

pril6 chname(
    pid32      pid,          /* ID of process to change */
    pril6      newprio       /* New priority */
)
{
    intmask mask;           /* Saved interrupt mask */
    struct procent *prptr;   /* Ptr to process's table entry */
    pril6 oldprio;          /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return (pril6) SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    kprintf("\n\nNew Syscall is easy\n\n");

    return oldprio;
}
```

File prototypes.h



```
extern void arp_ntoh(struct arppacket *);
extern void arp_hton(struct arppacket *);

/* in file ascdatetime.c */
extern status ascdatetime(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file cname.c */
extern pri16 cname(pid32, pri16);

/* in file clkupdate.S */
extern uint32 clkcount(void);

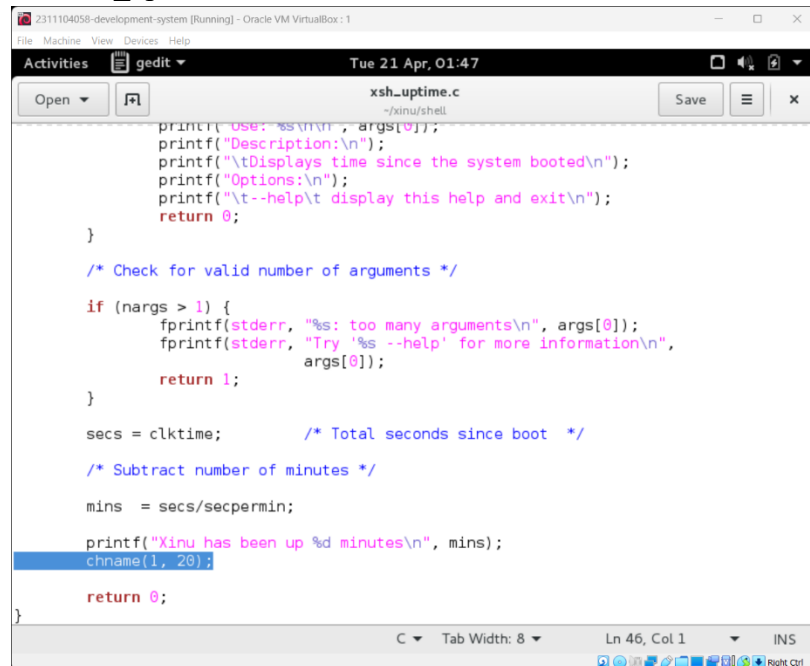
/* in file clkhandler.c */
extern interrupt clkhandler(void);

/* in file clkinit.c */
extern void clkinit(void);

/* in file clkdisp.S */
extern void clkdisp(void);

/* in file close.c */
```

File xsh_uptime.c



```
printf("Use: %s\n", args[0]);
printf("Description:\n");
printf("\tDisplays time since the system booted\n");
printf("Options:\n");
printf("\t--help\t display this help and exit\n");
return 0;
}

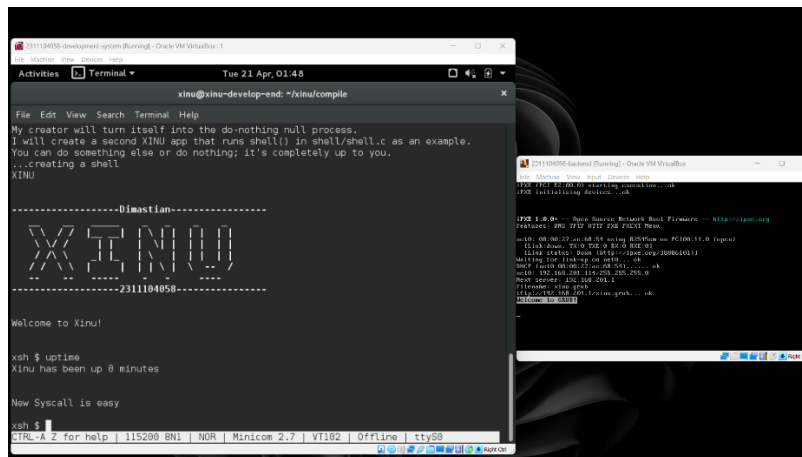
/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

secs = clktime; /* Total seconds since boot */

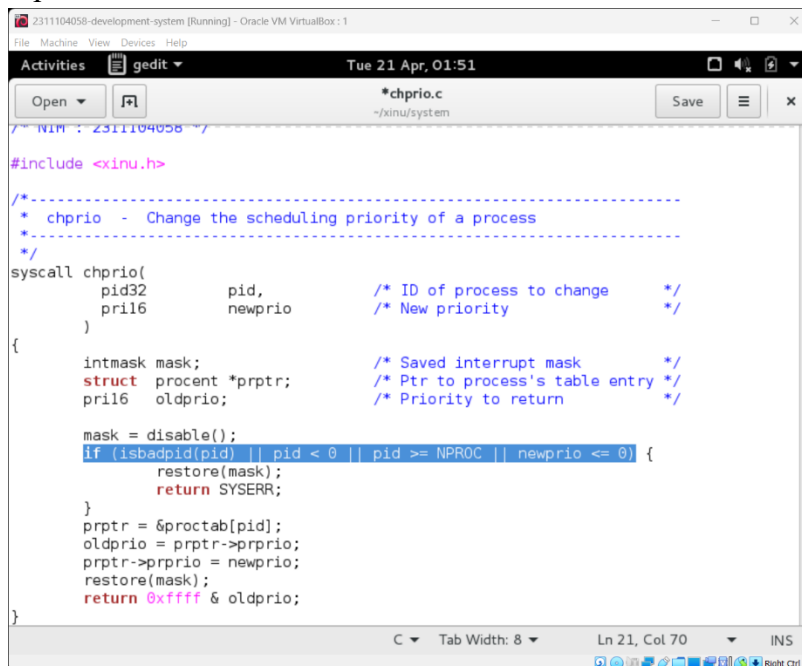
/* Subtract number of minutes */
mins = secs/secpermin;

printf("Xinu has been up %d minutes\n", mins);
cname(1, 20);

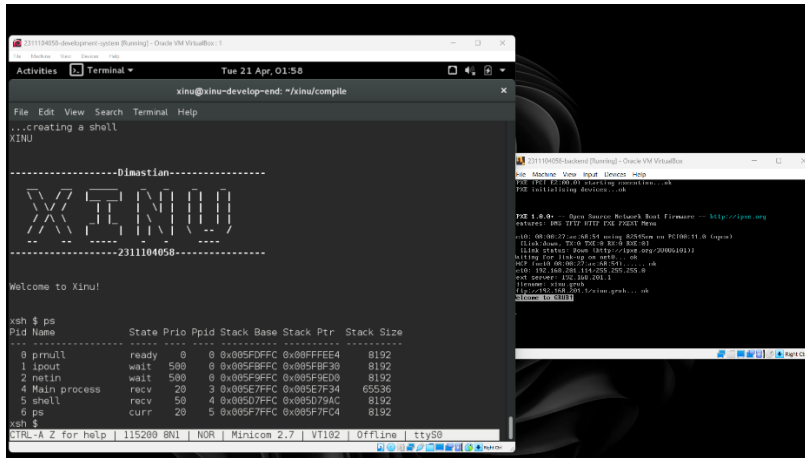
return 0;
}
```



2. Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input!



Sebelum menjalankan perintah uptime



```
xinu@xinu-develop-end:~/xinu/compile
...creating a shell
XINU

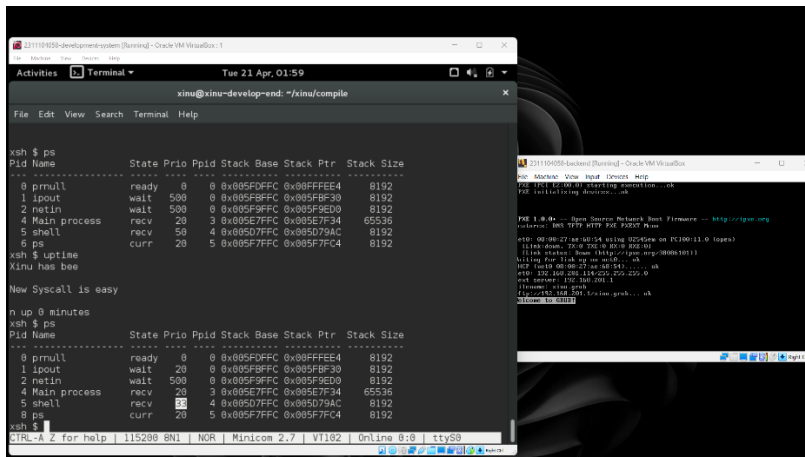
-----Dimastian-----
XINU
231104058-----

Welcome to XINU!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE04 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 natin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process  rcv    20  3 0x005E7FFC 0x005E7F34 65536
5 shell       rcv    50  4 0x005D7FFC 0x005D79AC 8192
6 ps         curr   20  5 0x005F7FFC 0x005F7FC4 8192

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Setelah menjalankan perintah uptime

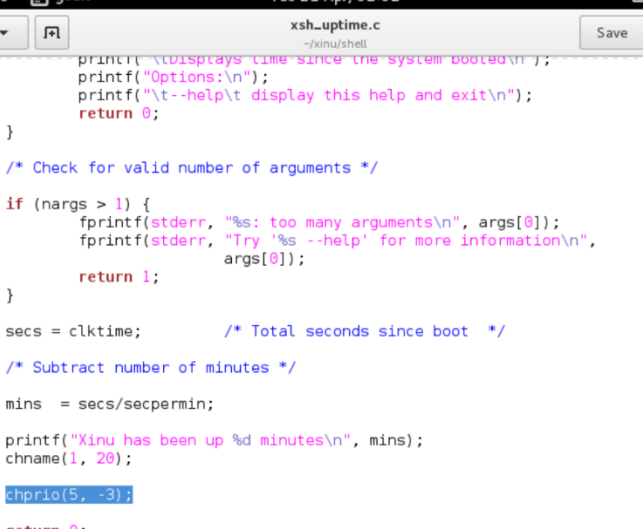


```
xinu@xinu-develop-end:~/xinu/compile

New Syscall is easy
n up 0 minutes
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE04 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 natin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process  rcv    20  3 0x005E7FFC 0x005E7F34 65536
5 shell       rcv    50  4 0x005D7FFC 0x005D79AC 8192
6 ps         curr   20  5 0x005F7FFC 0x005F7FC4 8192

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

3. Testing chprio syscall yang telah diubah
chprio(5,-3)



```
2311104058-development-system [Running] - Oracle VM VirtualBox: x
File Machine View Devices Help
Activities gedit Tue 21 Apr, 02:02
xsh_uptime.c
~/xinu/shell
Save
Open
Print
xsh_uptime.c
printf( "\tdisplays time since the system booted\n" );
printf( "Options:\n" );
printf( "\t--help\t display this help and exit\n" );
return 0;
}

/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

secs = clktime;          /* Total seconds since boot */

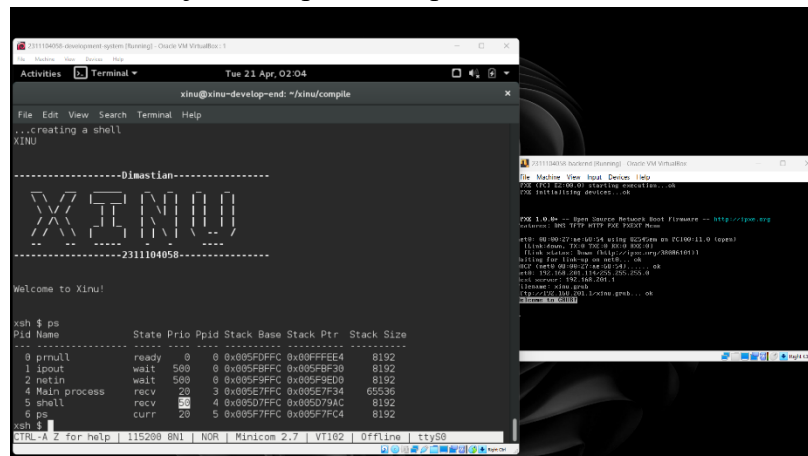
/* Subtract number of minutes */

mins = secs/secpermin;

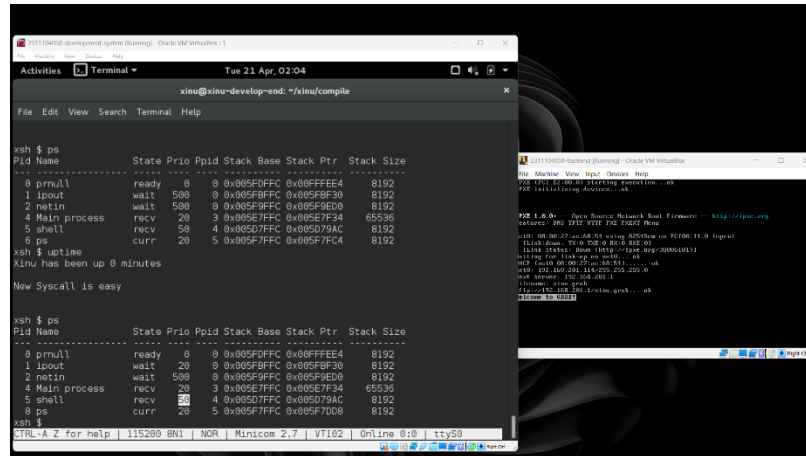
printf("Xinu has been up %d minutes\n", mins);
chname(1, 20);
chprio(5, -3);

return 0;
}
```

Sebelum menjalankan perintah uptime



Setelah menjalankan perintah uptime



```
xinu@xinu-develop-end:~/xinu/compile
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnall      ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005FAFFC 0x005F9ED0 8192
4 Main process  rcv    20  3 0x005E7FFC 0x005E7F34 65536
5 shell       rcv    50  4 0x005D7FFC 0x005D7AC 8192
6 ps         curr   20  5 0x00577FFC 0x0057FC4 8192

xsh $ uptime
Xinu has been up 0 minutes

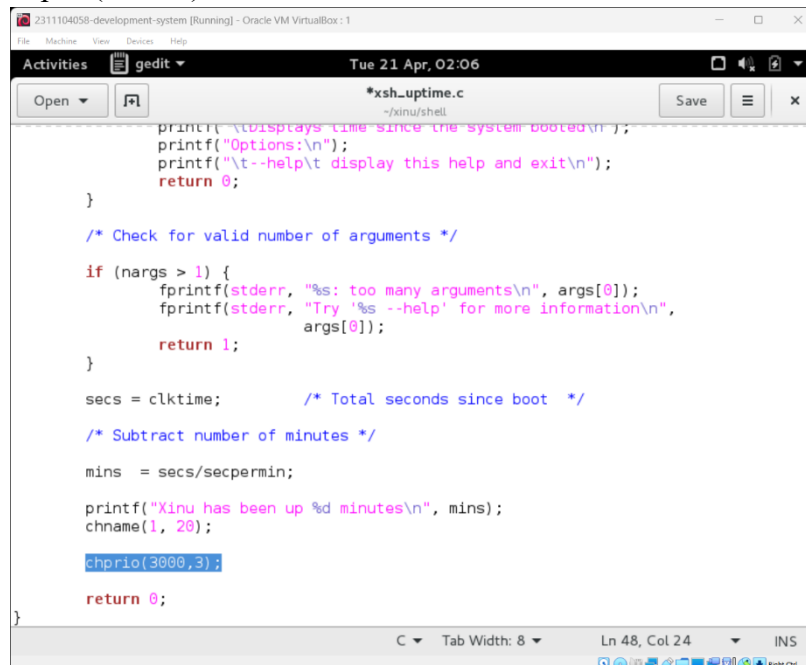
New Syscall is easy

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnall      ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   20  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005FAFFC 0x005F9ED0 8192
4 Main process  rcv    20  3 0x005E7FFC 0x005E7F34 65536
5 shell       rcv    20  4 0x005D7FFC 0x005D7AC 8192
6 ps         curr   20  5 0x00577FFC 0x0057F08 8192

xsh $
```

Saat uptime memanggil syscall ini, fungsi chprio di dalam kernel akan mengecek nilai newprio. Karena terdapat syarat newprio ≤ 0 akan me-return SYSERR (error), maka sistem langsung menolak permintaan tersebut. Sistem tidak akan menjalankan baris kode yang mengubah prioritas (prpr->prprio = newprio;). Yang menghasilkan Prioritas ID 5 aman dan tetap di angka 50.

chprio(3000,3)



```
*xsh_uptime.c
~/xinu/shell

printf("\nDisplays time since the system booted\n");
printf("Options:\n");
printf("\t--help\t display this help and exit\n");
return 0;

}

/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

secs = clktime;          /* Total seconds since boot */

/* Subtract number of minutes */

mins = secs/secpermin;

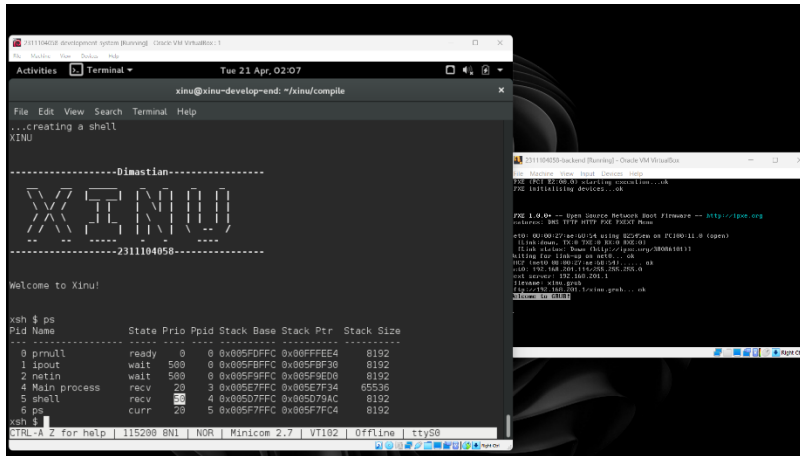
printf("Xinu has been up %d minutes\n", mins);
chname(1, 20);

chprio(3000,3);

return 0;

}
```

Sebelum menjalankan perintah uptime



```
xinu@xinu-develop-end: ~/xinu/compile
...creating a shell
XINU

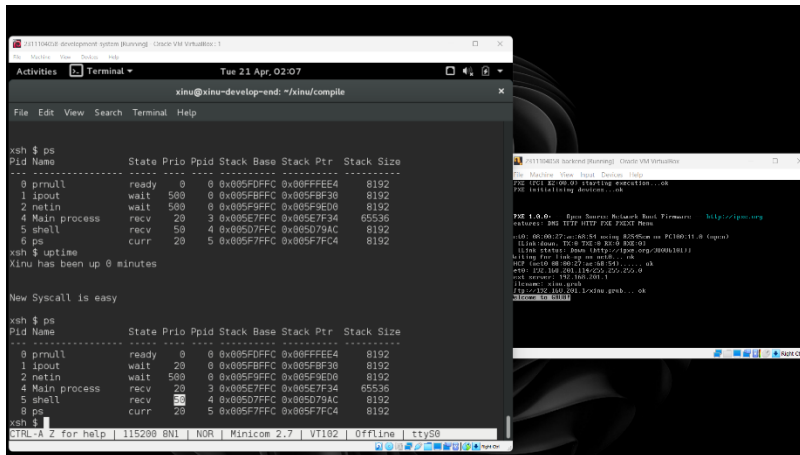
-----Dimastian-----
XINU
2311104058

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull      ready  0  0 0x005FDFFC 0x00FFFE4  8192
1 ipout       wait   500 0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500 0 0x005F9FFC 0x005F9E00 8192
4 Main process  rcv    20 3 0x005E7FFC 0x005E7F34 65536
5 shail       rcv    50 4 0x005D7FFC 0x005D79AC 8192
6 ps         curr   20 5 0x005F7FFC 0x005F7FC4 8192

xsh $
```

Setelah menjalankan perintah uptime



```
xinu@xinu-develop-end: ~/xinu/compile

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull      ready  0  0 0x005FDFFC 0x00FFFE4  8192
1 ipout       wait   500 0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500 0 0x005F9FFC 0x005F9E00 8192
4 Main process  rcv    20 3 0x005E7FFC 0x005E7F34 65536
5 shail       rcv    50 4 0x005D7FFC 0x005D79AC 8192
6 ps         curr   20 5 0x005F7FFC 0x005F7FC4 8192

xsh $ uptime
Xinu has been up 0 minutes

New Syscall is easy

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull      ready  0  0 0x005FDFFC 0x00FFFE4  8192
1 ipout       wait   500 0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500 0 0x005F9FFC 0x005F9E00 8192
4 Main process  rcv    20 3 0x005E7FFC 0x005E7F34 65536
5 shail       rcv    50 4 0x005D7FFC 0x005D79AC 8192
6 ps         curr   20 5 0x005F7FFC 0x005F7FC4 8192

xsh $
```

Sama seperti sebelumnya sistem memeriksa ID proses. Karena batas maksimal proses (NPROC) biasanya jauh di bawah 3000, kondisi $\text{pid} \geq \text{NPROC}$ terpenuhi dan sistem kembali me-return SYSERR. Permintaan dibatalkan secara diam-diam, sehingga tidak ada yang berubah dan sistem operasi Xinu tidak crash.